

Course Notes 7 - Homework Solutions

Problem 3-7

Solution:

The states cycle through 16 intervals (0000 to 1111). We partition the 80 seconds into 16 intervals of 5 seconds each.

- GNS (Green North/South) is on for 30s (6 intervals).
- YNS (Yellow North/South) is on for 5s (1 interval).
- RNS (Red North/South) is on for 45s (9 intervals).
- REW (Red East/West) is on for 45s (9 intervals). (Overlaps with RNS for 5s, 1 interval).
- GEW (Green East/West) is on for 30s (6 intervals).
- YEW (Yellow East/West) is on for 5s (1 interval).

Using K-maps or truth tables for the inputs A, B, C, D to map to the 6 light outputs, we can minimize the logic to AND/OR gates and inverters.

Problem 3-8

Solution:

A 3-bit input $X = X_2X_1X_0$ represents values from 0 to 7. The output $Y = X^2$ can range from 0 to 49 (requiring 6 bits $Y_5Y_4Y_3Y_2Y_1Y_0$).

Truth table maps:

- 000 -> 000000
- 001 -> 000001
- 010 -> 000100
- 011 -> 001001
- 100 -> 010000
- 101 -> 011001
- 110 -> 100100
- 111 -> 110001

Boolean equations can be simplified from this truth table for each output bit Y_i .

Problem 3-11

Solution:

Inputs: PS, LS, RS, RR

Outputs: PL, LL, RL

Operation rules mapped to logic:

1. $PL = PS$
2. $LL = \overline{PS} \cdot \overline{RS} \cdot LS + \overline{PS} \cdot LS \cdot RS \cdot RR$
3. $RL = \overline{PS} \cdot \overline{LS} \cdot RS + \overline{PS} \cdot LS \cdot RS \cdot \overline{RR}$

Problem 3-13

Solution:

Given equations:

$$F = A(C\overline{E} + DE) + \overline{A}D$$

$$G = B(C\overline{E} + DE) + \overline{B}C$$

Let $H(X, Y, Z) = X(C\overline{E} + DE)$. We can map the instances of the hierarchical component to formulate the overall schematic block diagram using three copies of the component.

Problem 3-14

Solution:

Hierarchical component: $H = \overline{X}Y + XZ$

Using Shannon's Expansion on $G = \overline{A}BC + \overline{A}BD + A\overline{B}C + A\overline{B}D$:

Evaluate G at $A = 0$ and $A = 1$, configuring inputs to the H block.

Problem 3-16

Solution:

Technology mapping to NAND gates for Figure 3-54. Replace each AND, OR, and NOT gate with their

NAND equivalents (e.g., an OR gate is a NAND gate with inverted inputs, which can be optimized with neighboring NANDs to cancel out double inversions).

Problem 3-27

Solution:

Inputs: S_i (sensors), M (master), A (alarm), L (lights), P (police).

Outputs: A_{out} , L_{out} , V_{out} , C_{out} .

Intrusion detected condition: $I = \overline{S_0} + \overline{S_1} + \overline{S_2} + \overline{S_3} + \overline{S_4} + \overline{S_5}$.

If $(I \cdot \overline{M}) == 1$:

$$A_{out} = A$$

$$L_{out} = L$$

$$V_{out} = 1$$

$$C_{out} = P$$

Else outputs are disabled (0).

Problem 3-28

Solution:

A 4-to-16 line decoder uses a most significant bit (MSB) to select between two 3-to-8 line decoders. The outputs of the 3-to-8 decoders are gated with the MSB (or its complement) using the 16 2-input AND gates.

Problem 3-29

Solution:

A 4-to-16 line decoder using five 2-to-4 line decoders with enable.

- The first 2-to-4 decoder takes the two MSBs and its 4 outputs connect to the enable pins of the remaining four 2-to-4 decoders.
- The other four 2-to-4 decoders share the two LSBs as their inputs.

Problem 3-37

Solution:

(a) 8-to-1 multiplexer using 3-to-8 decoder and 8x2 AND-OR: The 3 selection lines feed into the 3-to-8

decoder. The 8 outputs of the decoder are each ANDed with the corresponding 8 data input lines, and all are summed together in an 8-input OR gate.

Problem 3-44

Solution:

$$F_1 = \overline{X} + \overline{Z} + XYZ$$

$$F_2 = \overline{X} + \overline{Z} + \overline{X}YZ$$

$$F_3 = \overline{XYZ} + \overline{X} + \overline{Z}$$

Use a 3-to-8 line decoder for variables X, Y, Z. Convert equations to sum of minterms and attach corresponding decoder outputs to an external OR gate for each function.

Problem 3-47

Solution:

$$F(A, B, C, D) = \Sigma m(1, 3, 4, 11, 12, 13, 14, 15)$$

Using a 4-to-1 MUX with A, B as selection lines ($S_1 = A, S_0 = B$).

- $AB = 00: F(0, 0, C, D) = \Sigma m(1, 3) = \overline{C}D + CD = D$
- $AB = 01: F(0, 1, C, D) = \Sigma m(4) = \overline{C}\overline{D}$
- $AB = 10: F(1, 0, C, D) = \Sigma m(11) = CD$
- $AB = 11: F(1, 1, C, D) = \Sigma m(12, 13, 14, 15) = 1$

These four expressions feed into the I_0, I_1, I_2, I_3 lines of the MUX.

Problem 3-50

Solution:

Verify the circuit in Figure 3-58 implements a full adder:

Trace the Boolean logic for outputs S_i and C_{i+1} . By analyzing the XOR and NAND/AND configurations, the results will reduce to $S_i = A_i \oplus B_i \oplus C_i$ and $C_{i+1} = A_i B_i + C_i (A_i \oplus B_i)$, which are standard full adder equations.

Problem 3-51

Solution:

Obtain 1s and 2s complements:

- $10011100 : 1s = 01100011, 2s = 01100100$
- $10011101 : 1s = 01100010, 2s = 01100011$
- $10101000 : 1s = 01010111, 2s = 01011000$
- $00000000 : 1s = 11111111, 2s = 00000000$ (carry thrown away)
- $10000000 : 1s = 01111111, 2s = 10000000$

Problem 3-52

Solution:

Subtraction with 2s complement of subtrahend ($M - N = M + 2s_comp(N)$):

(a) $11010 - 10001 \rightarrow 11010 + 01111 = 101001 \rightarrow \text{Result: } 01001$

(b) $11110 - 01110 \rightarrow 11110 + 10010 = 110000 \rightarrow \text{Result: } 10000$

(c) $1111110 - 1111110 \rightarrow 1111110 + 0000010 = 10000000 \rightarrow \text{Result: } 0000000$

(d) $101001 - 000101 \rightarrow 101001 + 111011 = 1100100 \rightarrow \text{Result: } 100100$

Problem 3-59

Solution:

Comparisons A, B (unsigned). Output $X = 1$ if $A < B$.

This is essentially a borrow-out calculation for a subtraction $A - B$. We can implement it using subtractor logic where the final Borrow-Out signal equates to X . Alternatively, define standard magnitude comparator equations for $A < B$.